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Roy Posmanik of the Technion gave a talk called From Waste to Energy and Beyond, built around three questions: what actually makes something waste, what energy can be recovered from it, and what else the recovered material is good for.

His opening argument was definitional. Waste is not a property of a material. In a system that runs in one direction from extraction to disposal, waste is not an accident, it is the expected outcome, so the fix is not better disposal but a different system. He set this against the nine planetary boundaries and pointed out that several of the categories where we sit furthest outside the safe operating space are ones we push through consumption and disposal, including novel entities such as plastics and the nitrogen and phosphorus flows that come from how we handle nutrient-rich waste streams. His reframing was to stop managing waste as an endpoint and start engineering it as a resource, which means looking at a trash stream as a composition rather than a category. Carbon can become fuel, chemicals or carbon materials, the nitrogen, phosphorus and potassium can go back to agriculture, and water and synthetic polymers can be recovered separately. The example he used was the coffee capsule, which has an aluminum or polymer body, spent grounds, and a thin barrier lid, each of which has a different destination.

The energy half was hydrothermal processing in subcritical water, below the 374 degree critical point and under enough pressure to keep it liquid. Two things shift at once. The dielectric constant drops, so the water behaves less like a polar solvent and more like an organic one, while the dissociation constant rises, so the water itself supplies ions and the reaction becomes autocatalytic. Biomass goes in with high oxygen content and low energy density and comes out as hydrochar with much of the oxygen stripped, alongside an aqueous phase, an oil phase, and gas that is mostly carbon dioxide.

A postdoc in his group ran nine combinations of temperature and residence time on waste activated sludge and collapsed them into a single severity factor. As severity rose the oxygen to carbon ratio fell and the energy quality rose, which the combustion behavior confirmed. The detail that carried the rest of the talk was structural: under the electron microscope, low severity char kept a filamentous fibrous structure while high severity char turned amorphous. That is why the same process supports uses beyond fuel, including soil amendment, water holding capacity, and serving as a carrier for catalysts.

The phosphorus work was the strongest section. Nitrogen can be pulled from the atmosphere, but phosphorus comes from a finite mineral reserve, and meanwhile wastewater and sludge are full of phosphorus that we pay to remove as a pollutant. About 95% of it stayed in the hydrochar, but he was careful that recovery is not the same as function. Fractionation put most of it in the acid extractable fraction, which agronomy would call unavailable, and yet fertilization trials matched or beat triple superphosphate on total uptake. The explanation was polyphosphate, orthophosphate condensed into chains, which behaves as a slow release fertilizer.

The techno-economic piece was honest about limits. Scaled to a plant handling 21,000 tons of sludge a year and producing about 3,000 tons of hydrochar, phosphorus alone does not pay for the process. Roughly two thirds of the retained carbon proved stable, giving around 800 tons a year of creditable sequestration, and volume reduction avoids transport cost on top of that. His sensitivity analysis over capital subsidy and carbon prices of \$35, \$70 and \$140 per ton showed that the process fails outright with no support and can work with it. His point was that a waste project only pencils when you value the whole system rather than any single product.

He closed on the process water, which most people would treat as an effluent. Extract from hydrothermally treated spent coffee grounds, applied to the soil, primed tomato plants to defend themselves. In the leaf disc assay the extract alone did little and an elicitor alone did some, but extract followed by the elicitor produced a much larger oxidative burst. In the greenhouse, disease was lower even though the extract never touched the pathogen, which makes it systemic resistance rather than a pesticide effect. What I took from the seminar is that the real argument was about valuation, since the chemistry recovers several things at once and none of them pays on its own.